



Multilevel and multiobjective optimization in multidisciplinary design

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MULTILEVEL AND MULTIOBJECTIVE OPTIMIZATION IN MULTIDISCIPLINARY DESIGN

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Abstract

Multilevel nonlinear optimization problems arise in the design of complex systems. We discuss a class of multilevel methods for large problems with arbitrary couplings and propose a class of methods for solving the general multilevel optimization problem. The class can be used to regularize the multiobjective optimization problem. These approaches use the trust-region globalization strategy to achieve robust behavior.

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 AMS(MOS) subject classification: 65K05, 49D37

Introduction

The goal of this paper is to introduce the reader to the trust-region methodology for solving nonlinear programming problems, to demonstrate this methodology on a class of multilevel methods for solving constrained optimization problems, and to develop a generalization of this class to a general multilevel programming algorithm for use in multidisciplinary multiobjective optimization. This paper is limited to algorithmic discussions and an illustrative example. The analysis of the proposed general multilevel algorithm can be found in [2].

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Multidisciplinary optimization gives rise to large nonlinear programming problems that, as a rule, involve several, frequently competing, objectives. Aeronautical design may deal with maximizing performance while minimizing weight and drag, as just a few of the objective functions. The size and complexity of the problems necessitate decomposition into smaller subproblems. As a rule, the subproblems are partitioned along the lines of weaker couplings, processed separately, and then coordinated via another optimization problem. The new problem is frequently attacked in stages of a multilevel approach. Literature on multilevel methods for large multiobjective problems contains large volumes of references: [7], [8], [9], [10], [12], [17], [21], [22], [24], [23], [26], [28].

While practical multidisciplinary optimization has enjoyed success, multilevel formulations for a multidisciplinary optimization problem (MOP) are frequently subject to difficulties. Few formulations are clearly equivalent to the original problem, as mostly heuristics are used in creating a multilevel formulation. A related difficulty is that some of the problem couplings may be ignored in the decomposition. On the other hand, if the couplings are not ignored, some of the formulations suffer from lack of demonstrable robustness.

A general multiobjective (multicriteria, vector) nonlinear programming problem can be stated as follows:

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } C_i(x) = 0, \quad i = 1, \dots, m \\ &\quad \quad \quad g_i(x) \leq 0, \quad i = 1, \dots, p, \end{aligned}$$

where each $f_k(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ ($k = 1, \dots, M$) is a nonlinear smooth functional; feasibility means that the set of constraints is satisfied, and $C(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $g(x) : \mathbb{R}^n \rightarrow \mathbb{R}^p$, are smooth nonlinear functions.

Since vector optimization is not defined in the ordinary way, the problem may be stated as:

Find a feasible point $x_* \in \mathbb{R}^n$, which makes the components of $F(x) = \{f_k(x)\}$, $k = 1, \dots, M$, as

“preferable” as possible simultaneously.

Here a point x is preferable to a point y if $f(x) \leq f(y)$. However, other definitions of “preferable” may be more appropriate to some applications.

The multiobjective nature of the problem is now handled by numerous approaches. One class of approaches incorporates the components of the vector criteria into a single scalar objective (see, e.g., Wood [29]). Another class of approaches comprises various goal programming techniques, which can be characterized by the following: the most important objective serves as an optimization objective and the other objectives are constrained by setting bounds or “goals” for them. The problem is thus transformed into a nonlinear programming problem. Other methods, such as the multiobjective utility theory, are used.

As scalar notions of extremum do not apply to the vector problems, many approaches use the notion of Pareto optimality to achieve a balance between the objectives (see, for example, Sawaragi, Nakayama, and Tanino [19]). A point x_* is said to be Pareto optimal if and only if there is no point x such that $f_k(x) \leq f_k(x_*)$, $k = 1, \dots, M$, and $f_k(x) < f_k(x_*)$ for at least one k . In words, a point is Pareto optimal if one objective can be improved only by making other objectives worse.

This paper will discuss a class of algorithms for large optimization problems with arbitrary couplings, which serves as a foundation for a new algorithm for solving general multilevel problems. The general multilevel problem formulation may, in turn, be used as a method for formulating multiobjective optimization problems.

Our approach to multiobjective optimization is to restate the multiobjective problem as a multilevel optimization problem. This approach has not been extensively used due to both theoretical and practical difficulties of multilevel optimization.

The proposed method is grounded in the areas of trust-regions and multilevel methods for nonlinear programming ([1], [5], [4]). We acquaint the reader with these areas in the first two sections. We then propose and discuss an algorithm for general nonlinear multilevel optimization.

Trust-Region Preliminaries

The concepts of the trust-region methodology are easily explained on single-objective unconstrained problems, without loss of generality. Consider the following unconstrained minimization problem.

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } x \in \mathbb{R}^n, \end{aligned}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable.

Newton and quasi-Newton methods form a standard class of methods for finding local solutions to problem UNC. They can be summarized as follows:

1. Initialize;
2. Do until convergence:
 - Build a local model:

$$\phi_c(s) = f(x_c) + \nabla f(x_c)^T s + \frac{1}{2} s^T H_c s;$$
 Minimize $\phi_c(s)$ to obtain s_c ;
 - Set $x_+ = x_c + s_c$;
3. End.

Here x_c and x_+ denote the current and the next approximation to a solution, respectively, and H_c is the Hessian of f at x_c or an approximation. It is important to emphasize that the analysis imposes such mild assumptions on the second order information, that, for example, H_c can be 0, and global convergence is still assured. (See the Convergence Results section.)

Trust region algorithms are one of the major approaches used to improve the global behavior of such local model based algorithms. At each iteration, a typical algorithm finds a trial step by solving the following trust-region subproblem approximately:

$$\begin{aligned} &\text{minimize } f(x_c) + \nabla f(x_c)^T s + \frac{1}{2} s^T H_c s \\ &\text{subject to } \|s\| \leq \delta_c, \end{aligned}$$

where $\delta_c > 0$ is the trust-region radius, and $\|\cdot\|$ denotes the ℓ_2 norm. The idea is intuitive and natural. First, model the objective function in a region where the model is “trusted”. Second, accept the step if the model adequately predicts the behavior of the function or reject the step otherwise. Third, update the radius of the region in which the model is trusted.

If the step is accepted (see next section), that means that our estimate of the size of the region, δ_c in which the model can be trusted is reasonable. We can leave the δ_c the same for the next iteration or increase it, if the model performed exceptionally well. If the step is rejected, the size of the trust-region radius δ_c is decreased. There need not be a complicated rule for the adjustment of the trust-region radius. For instance it can be halved on decrease and doubled on increase.

The trust-region idea may be viewed as an adaptive move-limit strategy. However, unlike the move-limit strategy, the trust-region methods have rigorous and strong convergence properties.

Detailed treatment of the trust-region approach to unconstrained optimization and nonlinear equations can be found in [13], [14], [15], [16], [20], and [25]. In particular, [14] provides convergence results for the

class of trust-region methods, [13] and [15] provide encyclopedic discussions of the the method, while in the rest of the references practical and theoretical advances in the are introduced.

Trust-region algorithms have been used successfully to solve the general nonlinear constrained optimization problem. For example, the local step in the successive quadratic programming (SQP) method is found by computing a minimizer of the quadratic model of the Lagrangian at the current point, subject to linearized constraints. A trust-region algorithm based on SQP adds the trust-region constraint to the subproblem and additional constraints designed to ensure that the trust-region constraint and the linearized constraints are consistent. Extensive description of the trust-region approach to constrained optimization can be found in [18].

Measuring Progress

To evaluate a trial step, trust-region algorithms use merit functions, which are functionals related to the problem in a way that indicates improvement in the solution when the merit function is improved.

A natural choice for a merit function in problem UNC is the objective function itself. Let

$$\phi_c(s) = f(x_c) + \nabla f(x_c)^T s + \frac{1}{2} s^T H_c s$$

denote the quadratic model of the merit function. We define two related functions.

The **actual reduction** is defined as

$$ared_c(s_c) = f(x_c) - f(x_c + s_c),$$

and the **predicted reduction** is defined as

$$\begin{aligned} pred_c(s_c) &= \phi_c(0) - \phi_c(s_c) \\ &= -\nabla f(x_c)^T (s_c) - \frac{1}{2} s_c^T H_c s_c, \end{aligned}$$

so that the predicted reduction in the merit function is an approximation to the actual reduction in the merit function.

The standard way to evaluate the trial step in trust-region methods is to consider the ratio of the actual reduction to the predicted reduction, $\frac{ared_c(s_c)}{pred_c(s_c)}$. A value lower than a small predetermined value causes the step to be rejected. Otherwise the step is accepted. The agreement between the predicted and the actual reduction need not be large. For example, a typical value for the acceptable ratio between the two is 10^{-5} . In fact, the step may be accepted if a simple decrease in the objective function is attained. This is reasonable in applications with very expensive

function evaluations, where one would not wish to waste a computation. Thus, the approximation and the model do not have to agree well at all in order for the algorithm to yield satisfactory steps. The step must also satisfy an additional mild decrease condition called the Fraction of Cauchy Decrease. This condition is so mild that all methods of interest for computing the trust-region step satisfy it automatically.

For constrained optimization, the merit function is an expression that monitors the progress both in the objective function and the constraints. For instance, merit functions such as the ℓ_2 penalty function

$$P_1(x; \rho) = f(x) + \rho \|C(x)\|^2$$

can be used, where ρ is the penalty parameter. For discussions of merit functions for constrained optimization see, for example, [6] and [11].

Sufficient and Optimal Decrease

To assure global convergence of a trust-region algorithm for problem UNC, the trial step is required to satisfy a mild condition of decrease on the trial step, the **fraction of Cauchy decrease** (FCD) condition. (See [15] for a details.) It means that the trial step, s_c , must predict at least a fraction of the decrease predicted by the steepest descent step for the model within the trust region (the Cauchy step). I.e., for some fixed $\kappa_1 > 0$

$$pred(s_c) = \phi_c(s_c) - \phi_c(0) \leq \kappa_1 [\phi_c(s_c^{CP}) - \phi_c(0)],$$

where

$$s_c^{CP} = -\alpha_c^{CP} \nabla f(x_c) \text{ with } \alpha_c^{CP} = \begin{cases} \frac{\|\nabla f(x_c)\|^2}{\nabla f(x_c)^T H_c \nabla f(x_c)} & \text{if } \frac{\|\nabla f(x_c)\|^3}{\nabla f(x_c)^T H_c \nabla f(x_c)} \leq \delta_c \\ \frac{\delta_c}{\|\nabla f(x_c)\|} & \text{otherwise.} \end{cases}$$

Here s_c^{CP} denotes the step to the Cauchy Point described above. Also see [13].

A stronger condition, the **fraction of optimal decrease** property (FOD), allows one to prove stronger convergence results. A step s_c is said to satisfy FOD if it predicts at least a fraction of the decrease predicted by the optimal solution of the trust-region subproblem, i.e., for some fixed $\kappa_2 > 0$ we have

$$pred(s_c) = \phi_c(s_c) - \phi_c(0) \leq \kappa_2 [\phi_c(s_c^{OPT}) - \phi_c(0)],$$

where s_c^{OPT} solves the trust-region subproblem exactly.

The FCD condition is satisfied by all variants of the dogleg method ([13]) and by restricted subspace methods ([11]), for example. The stronger FOD condition is satisfied by most algorithms that attempt

to minimize accurately the local model on the trust region, for instance, by Levenberg-Marquardt type methods.

Convergence Results

Powell's global convergence theorem ([14]) for a trust-region algorithm shows the power of trust-region globalization ideas. (Here "globalization" means that a solution will be found from an arbitrary starting point; it does not mean that the global optimum will be found.) It states that if f is uniformly continuously differentiable and $\{H_i\}$ (Hessian approximations) are only assumed to be uniformly bounded, then the sequence of iterates generated by a FCD trust-region algorithm is well-defined and satisfies

$$\liminf_{i \rightarrow \infty} \|\nabla f(x_i)\| = 0.$$

In [25] stronger convergence results are shown for trust-region algorithms with steps that satisfy FOD. Specifically, if the Hessian is Lipschitz continuous, and if exact Hessians are used in the local models, then any limit point of the iterates satisfies second order necessary conditions, i.e., has a positive semidefinite Hessian. Furthermore, under some reasonable additional assumptions, the iteration sequence converges q-quadratically to a second order necessary point for UNC.

Detailed treatment of the unconstrained minimization theory and practice can be found in [15], [16], [25], and [20].

In this paper, the proposed methods will be addressed as "multilevel methods" or as "class of multilevel methods", because the methodology comprises an entire class of algorithms, depending on the way the optimization step is computed. The theory of the methods covers this wide range of the algorithms.

Multilevel Methods for Nonlinear Equations and Equality Constrained Optimization

The algorithms introduced here are based on the recently proposed class of multilevel algorithms for equality constrained optimization and nonlinear equations ([1]). Consider the following problem (EQC):

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } C(x) = 0, \end{aligned}$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $C: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $m \leq n$, are at least twice continuously differentiable, and m and n are assumed to be large. We consider the equality constraints here because the differential constraints of the multidisciplinary optimization problems may be

viewed as equality constraints. By differential constraints the analysis codes for the solution of partial or ordinary differential equations are meant. In addition, the computationally intensive part of the approach to inequality constrained optimization that is of interest to us is the equality constrained nonlinear programming problem.

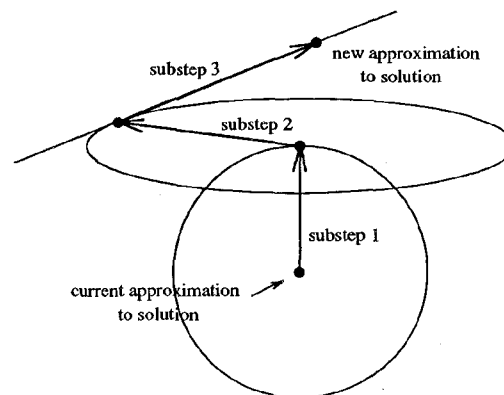


Figure 1. Multilevel Optimization Algorithm for Two Blocks of Constraints

We start by partitioning the constraint system, $C(x) = 0$ into M blocks. The problem can be written as

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } C_1(x) = 0 \\ &\quad C_2(x) = 0 \\ &\quad \vdots \\ &\quad C_M(x) = 0. \end{aligned}$$

The problem is clearly equivalent to the original one. In practice, the decomposition should be obvious. The discipline analyses in multidisciplinary optimization represent a naturally occurring small number of large blocks of constraints and, hence, a natural decomposition of the problem. To compute the trial step at the current approximation to the solution, we first find a substep, which is an approximate minimizer of the model of the first block of constraints in a restricted ellipsoidal region. We take the substep, and then find the second substep, which is an approximate minimizer of the model of the second block of constraints restricted to a hyperplane parallel to the null space of the Jacobian of the first block of constraints in a smaller dimensional ellipsoidal region. See Figure 1.

Restricting the model of each subsequent block of constraints—and then the model of the objective function—to the intersection of the hyperplanes described above ensures that the predicted improvement for that block is retained throughout the com-

putation of the total step. This is usually the feature that is missing in the existing multilevel algorithms - by processing a part of the problem without assuring rigorous preservation of the progress already made in another part of the computation may prevent convergence.

The multilevel class of methods introduced above is characterized by the following features:

- There is no assumption on the structure of the problem and no need to identify weakly-coupled subsystems. Thus, there is no separability assumption. Once the constraints are partitioned, the algorithm then solves progressively smaller dimensional subproblems to arrive at the trial step.

- The multilevel methods belong to the class of out-of-core methods. To the author's knowledge, they form the only class of methods for general nonlinear programming that require access only to the currently processed block of constraints. Other constraint blocks do not need to be kept in memory. Thus, the methods are expected to be well suited to very large problems, as, theoretically, there is no limit to the size of the problem they can handle.

- The computed trial steps are required to satisfy very mild conditions, both theoretically and computationally. In fact, the contributing substeps can be computed by different optimization algorithms and are only required to satisfy a decrease condition and a boundedness condition, which are both satisfied in practice by most methods of interest. This feature is of great practical significance because in applications like MDO various constraint blocks may originate from different disciplines and will require different approaches to solving the subproblems.

- The algorithms have been proven to converge under reasonable assumptions.

The General Multilevel Formulation for Multiobjective Optimization

Notice that the multilevel algorithm for equality constrained optimization described earlier solves the following problem ([1]):

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } x \in \operatorname{argmin} \|C_M(x)\|^2 \\ &\dots \\ &\text{subject to } x \in \operatorname{argmin} \|C_1(x)\|^2, \end{aligned}$$

where "argmin" denotes the set of minima of a particular function.

In words, the notation means, "find a minimizer of f which also minimizes $\|C_M(x)\|^2$ while minimizing $\|C_{M-1}(x)\|^2$ and so on, while the minimizer of $\|C_2(x)\|^2$ also minimizes $\|C_1(x)\|^2$. This is a multilevel problem with a specific structure of the all

objectives except the top-level one. It is natural to attempt the extension of the problem to the general multilevel optimization problem (MLO).

Consider again the problem objectives $f_1, \dots, f_M$. The general multilevel optimization formulation is:

Problem MLO:

$$\begin{aligned} &\text{minimize } f_M(x) \\ &\text{subject to } x \in \operatorname{argmin} f_{M-1}(x) \\ &\dots \end{aligned}$$

$$\text{subject to } x \in \operatorname{argmin} f_1(x),$$

where $f_1, \dots, f_M : \mathbb{R}^n \rightarrow \mathbb{R}$ are sufficiently smooth.

Note that here only unconstrained general MLO is considered, since the extension to the constrained case is self-explanatory.

This formulation is not new. It is a general lexicographic formulation. However, the analogy with the multilevel formulation for problem EQC above, allows us to suggest that the algorithm analogous to the multilevel algorithm for problem EQC has a good chance of succeeding for the general problem. It is also natural to consider the application of the general problem to multiobjective optimization.

Let $f_1, \dots, f_M$ be the problem objectives, arranged in the order of decreasing significance, i.e., f_M is the least important objective, while f_1 is the most important objective. Note that this significance does not need to be quantified in any way other than the establishment of the order. Then the MLO formulation makes the problem of minimizing a number of unconstrained functional simultaneously a constrained multilevel problem, with the least important objective serving as an objective function, while the rest of the objectives are transformed into constraints which are themselves constrained minimization problems. Only the most important objective forms an unconstrained minimization problem. Again, we consider only the unconstrained objectives, because the extension to the constrained case is contained in the unconstrained formulation.

Note that although the problem seems to be a nested problem, it is only the formulation that is nested. The algorithm proposed for solving the problem will not proceed in the obvious way, i.e., it will not solve the innermost problem to completion, the proceed to the next one, etc.

The MLO formulation gives us a way to "regularize", i.e., to make well-posed, the ill-defined problem of multicriteria optimization. Here engineering insight will enter into defining the order in which the optimization levels are stated. The reader will see that if the objective values of all the objectives, except the most important one, were known at a solution to an MLO problem, then one would have a goal

program. Thus, it is possible to view MLO as related to goal programming except that the order of importance of the objectives needs to be specified rather than goals for each objective value.

Existing work in multilevel and, in particular, bilevel optimization (see [27] for a review) deals only with functions under extremely strong assumptions of convexity—and sometimes separability—and has many theoretical difficulties. Typically, in applications the convexity assumptions are not met, while the separability assumptions may not always be made, especially in highly coupled problems of multidisciplinary optimization.

In recent work (see [1], [5], [4]), algorithms for multilevel optimization of problems with special structure have been shown to exhibit global convergence under reasonable assumptions. The structure was that of a single objective nonlinear programming problem.

The multilevel approach to multiobjective optimization is an extension of the multilevel algorithms in [1] and it arrives from the current approximation of the solution to the next approximation to the solution by solving a sequence of optimization subproblems each of which will serve to set a “goal” in defining the region of sufficient decrease in the merit function for the final optimization subproblem. We have a new, promising, merit function that will allow us to evaluate the progress of the algorithm toward a solution.

The multilevel formulation is sequential because we make no assumption on separability and do not decouple the problem. If the problem is separable or partially separable, i.e., not fully coupled, then the formulation produces parallelism in an obvious way. In particular, the engineering decomposition formulations become a special case ([3]). This is of interest, because the existing decomposition/integration formulations lack theoretical foundation and are not robust in practice. We suggest that the general multilevel optimization analysis will provide the necessary foundation and suggestions for improving the robustness of existing engineering multilevel algorithms. ([3]).

The Algorithm

The main idea is based on achieving staged successive decrease conditions for all the objectives, which is a direct extension of the multilevel ideas for problem EQC.

The algorithm we propose for MLO can be stated as follows:

Compute the trial step for problem MLO to produce an FCD or FOD on the quadratic model of f_M subject

to producing FCD or FOD on the quadratic model of f_{M-1} , etc., subject to producing FCD or FOD on the quadratic model of f_1 .

In practice, the algorithm would be implemented in the following way. The inner problem would be solved by a conventional trust-region approach to unconstrained minimization to produce the FCD or FOD “goal” for the quadratic model of f_1 about the current point. The second problem would be solved on the model of f_2 , built about the new point, in the appropriate null space to preserve the predicted decrease on the model of f_1 . And so on. In other words, step 1 of the algorithm will minimize approximately

$$m_1(s) = f_1(x_c) + \nabla f_1(x_c)^T s + \frac{1}{2} s^T H_1(x_c) s$$

with $\|s\| \leq \delta_1$ to produce s_1 .

Step 2 would minimize approximately

$$\begin{aligned} \text{minimize } m_2(s) = & f_2(x_c + s_1) + \nabla f_2(x_c + s_1)^T s + \frac{1}{2} s^T H_2(x_c + s_1) s \\ \text{subject to } & (\nabla f_1(x_c) + H_1(x_c) s_1)^T s = 0, \text{ and} \\ & \|s\| \leq \delta_2. \end{aligned}$$

Step 3 will approximately minimize the model of f_3 about the new point subject now to the two subspace restrictions and its own trust-region constraint.

To account for the separate processing of the functions, we use the following merit function:

$$P(x; \rho_1, \dots, \rho_{M-1}) = f_M + \rho_{M-1}(f_{M-1} + \rho_{M-2}(\dots + \rho_1 f_1)) \dots,$$

where $\rho_i, i = 1, \dots, M-1$ are the penalty parameters. The merit function is used to evaluate the progress of the algorithm toward a solution in the usual manner of comparing the merit function at the current point and the new point. The penalty parameters are used in the following way. Minimizing the model of f_1 from the initial point is guaranteed by analysis to predict decrease in the model of the first function. However, we do not know what the first step does to the rest of the objectives. The first penalty parameter balances the predicted decrease in the first objective with the possible increase that the first substep produces in the second objective. The second penalty parameter balances the common predicted decrease in the first and the balanced second objectives with the possible predicted increase in the third objective, and so on. The result of the adjustment of the penalty parameters is the predicted reduction for the entire merit function.

Clearly, if the extremal values of all the objectives except the most important one were known at a solution to an MLO problem, then one would have a goal

x_1	x_2	$f_1(x_1, x_2)$	$f_2(x_1, x_2)$
2	4	5	58
3	5	7	40
4	7	18	19
5	8	30	11
6	10	57	6

Table 1: Pareto-optimal solutions for a 2x2 example.

program. One can think of our algorithm as a way to set goals adaptively for each iteration.

Given standard assumptions on the functions (continuity, regularity), the algorithm can be shown to converge to a Pareto optimal point ([2]). It remains to be shown whether the standard assumptions are too strong for practical problems and how far they can be relaxed.

Numerical Results

A small set of bilevel problems was solved by the bilevel version of the algorithm. In particular, the algorithm was applied to problems with two quadratic objective functions. For example, consider the following two objectives ([12]):

$$\begin{aligned} f_1(x_1, x_2) &= (x_1 - 2)^2 + (x_2 - 4)^2 + 5 \\ f_2(x_1, x_2) &= (x_1 - 6)^2 + (x_2 - 10)^2 + 6. \end{aligned}$$

The unconstrained minima of f_1 is (2,4) and of f_2 - (6,10), and some of the Pareto optimal points are shown on Figure 2 and the Table.

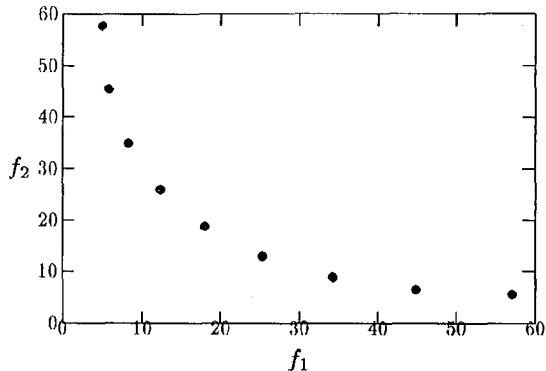


Figure 2. A Numerical Example

A point of the Pareto optimal set was always found. It appears that the starting point was always sufficiently close to the solution to guarantee fast rates of convergence. However, there is a number of unresolved questions. Without additional consideration, the algorithm converges either to the point (2,4) or

to the point (6,10), depending on whether f_1 or f_2 is the innermost objective. This is reasonable, given the statement of the problem and the algorithm. However, the numerical example shown is not very interesting as a multilevel problem because both functions are convex, hence, strictly speaking, the minimizer of each objective is the solution of the multilevel problem. The problem would be truly multilevel if the innermost objective, f_1 or f_2 had a continuum of minimizers, which is a likely case.

The solutions (2,4) and (6,10) place overwhelming significance on either objective f_1 or objective f_2 . This may be appropriate for the physical properties of the problem. More likely, the user may be more interested in a more balanced solution. There are several ways to achieve a more balanced solution in a multilevel approach. One may consider scaling the innermost function or functions, so that less emphasis is placed on them. The approach that is being considered is instead to modify the stopping criterion for the algorithm. Once the "boundary" Pareto point is obtained, a number of exploratory steps ([2]) to test nearby points. In particular, for the example above, exploratory optimization found the points shown on Figure 2 (to some precision). Exploratory optimization steps are being considered now for rigorous inclusion in the algorithm.

The next step in testing is to attempt solution of larger problems, and in the case of success, of realistic problems.

Concluding Remarks

We proposed an approach to bilevel and multilevel optimization, which regularizes the ill-posed problem of multiobjective optimization. The approach requires the user to indicate the order of importance for the objectives, but does not require the importance to be quantified in any other way.

The main difficulties of the multilevel formulations have always been the possible intractability of the feasible set for the problem and in showing the existence of search directions under reasonable assumptions. We conjectured and proved (see proof in [2]) several theoretical properties of the formulation and the algorithm that are designed to ensure robustness.

The current and future work is concerned with a number of practical and theoretical issues. For example, the algorithm is well-posed when the number of objectives is smaller than or equal to the number of optimization variables. A possible resolution for the case of more objectives than variables is to treat the $M = n$ objectives via the multilevel approach and treat the rest of the objectives via a goal

programming-like approach. Another promising approach under consideration is more similar to goal programming and does not restrict the number of objectives.

Visualization strategies that will assist in the choice of Pareto optimal points are being considered as well.

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